Special instructions for invigilators: None.

Information for Candidates:

Numbers in brackets in the right margin (e.g. [5]) indicate maximum marks for each section of each question.

The following constants may be used:

electron charge : $e = 1.6 \times 10^{-19}$ C

permittivity of free space : $\varepsilon_o = 8.85 \times 10^{-12}$ F/m

relative permittivity of silicon : $\varepsilon_r = 12$

Planck's constant : $h = 6.63 \times 10^{-34}$ J s

Boltzmann's constant : $k = 1.38 \times 10^{-23}$ J/K

speed of light : $c = 3.0 \times 10^{8}$ m/s

The eigenvalue equations for TE modes in a symmetric slab waveguide of thickness d are

$\kappa = k_{1x} \tan(k_{1x} d/2)$ and $\kappa = -k_{1x} \cot(k_{1x} d/2)$

Answers

1. You should attempt all parts of this question. Short answers only are required; there is no need to re-state the questions in your answer book, but you should show any calculations you use to arrive at your answers, state any assumptions or approximations made, and give a brief (one or two lines) explanation where appropriate. All parts have equal value.

[20]

a) Calculate the time needed for an optical pulse to travel through 250 km of a silica-based optical fibre.

Solution: Assuming an $n' = 1.5$, $u' = \frac{c}{n'} = 2 \cdot 10^8 m/s$

Therefore: $\tau = \frac{L}{u'} = \frac{25 \cdot 10^4}{2 \cdot 10^8} = 1.25\ ms$

b) A slab waveguide supports 3 transverse electric (TE) modes, m = 0, 1 and 2, for a nominal wavelength λ_o of 1.10 μm. At least how many modes could be supported for λ_o = 0.55 μm?

Solution: Since at 1.10 μm mode 2 is supported, at half the wavelength, modes up to 4 may be supported. Hence, at 0.55 μm the waveguide can support at least modes m = 0, 1, 2, 3 and 4.

c) An optical fibre with a silica core has a numerical aperture of 0.20. Calculate the refractive index difference Δn.

Solution: The Numerical Aperture is linked to the refractive indices of an optical fibre by the equation:

$$NA = \sqrt{n_1^2 - n_2^2}$$

Estimating for silica an $n' = 1.5$, and $n_1 + n_2 = 2 \cdot n'$, it will be:

$$NA = \sqrt{2 \cdot n' \cdot \Delta n}$$

Hence:

$$\Delta n = \frac{NA^2}{2n'} = \frac{0.04}{3} = 0.0133$$

d) Which type of noise in optical communications cannot be reduced by employing an optical amplifier? Provide a brief justification.

Answers

Solution:

The shot noise can't be reduced by an amplifier because any signal amplification will equally amplify the shot noise. In contrast, noise beyond the amplifier such as the receiver noise is reduced.

e) A photodiode produces 16 nA of photocurrent for an incident optical power of 33 nW at a nominal wavelength λ_o = 663 nm. Calculate the quantum efficiency η of the diode.

Solution:

Number of electrons = η times number of photons:

$$N = \frac{E_{tot}}{E_{phonon}} = \Phi \cdot \frac{t}{h \cdot \frac{c}{\lambda}} = \frac{\Phi t \lambda}{hc}$$

Then:

$$I_{photo} = \frac{\eta \cdot N \cdot e}{t} = \frac{\eta \Phi \lambda e}{hc}$$

Hence:

$$\eta = \frac{Ihc}{\Phi \lambda e} = \frac{6.63{\times}10^{-34} \times 3{\times}10^{8} \times 16{\times}10^{-9}}{1.6\ 10^{-19} \cdot 0.663 \cdot 10^{-6} \cdot 33 \cdot 10^{-9}} = 0.91$$

f) A certain symmetric slab waveguide of thickness d = 6 μm supports exactly 3 TE modes for λ = 750 nm. Calculate the range of possible values of numerical aperture (NA) of this waveguide.

Solution:

The radius R in the κd/2 vs $k_{1,x}d/2$ diagram is: $R = NA \cdot \pi \cdot d/\lambda$.

To support exactly 3 TE modes, the arc must cross exactly three tangential curves therefore, R must be between π and 3π/2. Hence:

$$\frac{\lambda}{d} \leq NA \leq \frac{3\lambda}{2d}$$

$$0.125 \leq NA \leq 0.1875$$

g) The wavelength of zero dispersion is different in silica fibre and in bulk silica glass. Provide a brief explanation.

Solution:

Answers

The chromatic dispersion for the fibre is a combination of the material dispersion of the glass, and waveguide dispersion, which is caused by the changing mode shape with wavelength, and consequently the changing distribution of the mode's power between the core and cladding. This latter effect shifts the dispersion zero.

h) In semiconductors, for small electric field E values, the drift velocity is proportional to E and another quantity. Provide the name and the units of that quantity. Is this quantity higher for electrons or for holes?

Solution:

This is the mobility. It is usually higher for electrons than holes.

i) A steady electromagnetic wave of 10^{11} photons per second at λ_o = 760 nm arrives at an optical receiver. Calculate the power level of the optical signal in both nW and dBm.

Solution:

Energy of each photon: $E = h \cdot v = h \cdot \frac{c}{\lambda} = 6.63 \cdot 10^{-34} \cdot 3 \cdot \frac{10^8}{0.76 \cdot 10^{-6}} = 26.17 \cdot 10^{-20}$ J

Power:

$$P = 10^{11} \cdot 26.17 \cdot 10^{-20} = 26.17\ nW$$

$$P_{dBm} = 10 \cdot \log \frac{(26.17 \cdot 10^{-9})}{10^{-3}} = -45.82\ dBm$$

j) The optical absorption of a particular glass material results in 8% intensity loss at a distance of 250 m. What is the attenuation coefficient of this material in dB/km.

Solution:

Loss in dB at 250 m.

$$10 \log \left(\frac{I_{out}}{I_{in}}\right) = 10log0.92 = -3.6\ dB$$

Therefore:

$$Attenuation = -3.6 \cdot 4 = -14.4\ dB/km$$

Answers

2.

a) The symmetric slab waveguide shown in Fig. 2.1 is operating at a given nominal wavelength λ_o. For the lowest order transverse electric (TE) mode (m=0), the electric field amplitude at the core boundary is found to be 0.5 times that at the centre, i.e. $E(x=d/2) = 0.5\ E(x=0)$. Calculate the value of $X = k_{1,x}\, d/2$ for this mode.

[4]

b) Calculate the number of TE modes supported by the waveguide of part (a). Plot a diagram of Y vs X (with $Y = Kd/2$), showing the supported modes.

[4]

c) Show that for the m=0 mode described in (a), the effective index n' satisfies the relation: $n'^2 = (3n_1^2 + n_2^2)/4$

[6]

d) If the m=0 mode has unit electric field amplitude (in V/m) at x=0, find the amplitude at $x = 2d$.

[6]

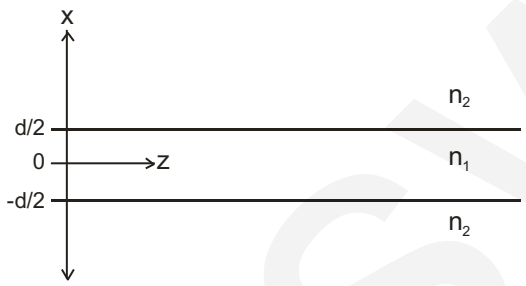


Figure 2.1 Slab waveguide

Solution:

a) In the core, for $m = 0$:

$$E(x) = A\cos(k_{1,x}x)$$

$$A\cos\left(\frac{k_{1,x}d}{2}\right) = 0.5 \cdot A\cos(0)$$

$$X = \frac{k_{1,x}d}{2} = \frac{\pi}{3}$$

b)

$$Y = X \cdot \tan X = \sqrt{3} \cdot \frac{\pi}{3}$$

Answers

$$R = \sqrt{\left(\frac{\pi}{3}\right)^2 + \left(\frac{\pi}{3}\right)^2 \cdot 3} = \frac{2\pi}{3}$$

The line can support modes 0 and 1.

c) The effective index is given by:

$$n' = \beta / k_0$$

Where $\beta^2 = n_1^2 \cdot k_0^2 - k_{1,x}^2$

$$R = NA \cdot k_0 \cdot d/2$$

$$d = \frac{4\pi}{3 \cdot NA \cdot k_0}$$

$$\frac{k_{1,x}}{k_0} = \frac{2\pi}{3dk_0} = \frac{2\pi}{3 \cdot \frac{4\pi}{3NA}} = \frac{NA}{2}$$

Now:

$$n' = \sqrt{n_1^2 - (k_{1,x}^2/k_0^2)}$$

$$n' = \sqrt{n_1^2 - NA^2/4} = \frac{\sqrt{3n_1^2 + n_2^2}}{2}$$

d) In the cladding:

$$E(x) = B \cdot e^{-kx}$$

Specifying B:

$$E\left(\frac{d}{2}\right) = \frac{1}{2}E(0) = 0.5$$

Answers

Hence:

$$B = \frac{1}{2}e^{\frac{kd}{2}}$$

Now:

$$E(2d) = \frac{1}{2}e^{\frac{kd}{2}} \cdot e^{-2kd} = \frac{1}{2}e^{-\frac{3kd}{2}} = \frac{1}{2}e^{-3Y} = \frac{1}{2}e^{-\sqrt{3}\pi} = 0.002167\ V/m$$

3. a) A certain Fabry-Perot laser diode has a cavity length of 750 μm and an effective index for the cavity modes of 3.35. The nominal wavelength is 1150 nm. Find the spacing between the nominal wavelengths of the longitudinal modes in nm.

[4]

b) For the laser in (a), estimate the number of longitudinal modes in the spectrum.

[4]

c) The laser is then modified by adding distributed Bragg reflectors. Calculate a suitable period Λ for these. What is the main advantage of such structures?

[4]

d) Using a quantum efficiency of 0.75, calculate the slope efficiency S of this laser. Hence, if a small sinusoidal current of amplitude 10 μA is added to a suitable bias current to bring the laser above threshold, what will be the amplitude of the sinusoidal component of the output optical power?

[4]

e) Explain why the maximum modulation rate of laser diodes can be much higher than that of an LED.

[4]

Solution:

a) The cavity length must be a multiple of $\lambda/2$, where λ is the wavelength in the cavity. If λ_0 is the free space wavelength, then:

$$L = \frac{m\lambda_0}{2n'}$$

$$\lambda_0 = \frac{2n'L}{m}$$

$$\frac{\Delta\lambda_0}{\Delta m} = -\frac{2n'L}{m^2} = -\frac{\lambda_0^2}{2n'L} = 0.26\ nm$$

b)

The photon energy at the nominal wavelength is $E = \frac{hc}{\lambda_0} = 1.08\ eV$

Answers

Thermal bandgap variations result in a variation of λ.

A thermal energy variation of $\Delta E = 2kT = 50\,meV$ is assumed. The resulting $\Delta\lambda$ can be calculated as follows:

$$E = h\nu \Rightarrow E = hc/\lambda$$
$$\frac{\Delta\lambda}{\Delta E} = -\frac{hc}{E^2} = -\frac{\lambda}{E}$$

$$\Delta\lambda = -\frac{\Delta E}{E}\lambda$$

Hence:

$$\Delta\lambda = -\frac{50}{1.08}\cdot 1.15 = 53.3\text{ nm}$$

Taking into account the mode spacing, the number of existing modes is:

$$\mathrm{N} = \frac{\Delta\lambda}{0.26\text{ nm}} = 205\text{ modes}$$

c)

The grating period Λ will need to be:

$$\Lambda = \frac{\lambda_0}{2n'} = \frac{1.15}{6.7} = 0.17\,\mu m$$

The grating provides wavelength-selective reflection. Thereby it promotes a single longitudinal mode, narrowing down the $\Delta\lambda$.

d)

The slope efficiency will be:

$$S = \frac{h\nu}{e}\eta = \frac{hc}{e\lambda_0}\eta = 0.81$$

The sinusoidal output power variation amplitude will then be:

$$P = S\cdot I = 8.1\,\mu W$$

e)

The LED speed is limited by the turn-off time, which is long, due to the spontaneous recombination of remaining carriers around the junctions, which prolongs the emission beyond the current switch off time.

In the laser, light still in the cavity causes rapid depletion of carriers by stimulated recombination.

Answers

4. a) Two optical links are constructed over the same length. Link A uses a receiver with noise equivalent power (NEP) of 12.0 pW/√Hz, and fibre with loss 0.45 dB/km. Link B uses a receiver with NEP of 9.0 pW/√Hz, and fibre with loss 0.47 dB/km. The transmitted power and bit rate are the same for both systems. The link capacity (maximum bit rate) is found to be limited by receiver noise, and to be the same, in both systems. Calculate the link length L, stating any approximations or assumptions made.

[6]

b) Estimate the bit rate in the links of part (a). Hence, show that for these links the shot noise will be significantly less than the receiver noise, assuming that the optical signal-to-noise ratio (SNR) equals 12 when considering receiver noise only.

[6]

c) If the optical receiver has a quantum efficiency of 1, prove that the shot noise and receiver noise will be equal if the received optical power equals the noise equivalent power squared divided by twice the photon energy.

[4]

d) Show that if an optical link is limited by shot noise, the number of electrons per bit in the photocurrent is simply equal to the square of the optical signal-to-noise ratio. State any assumptions or approximations made.

[4]

Solution:

a) The link A requires 12/9 more received power to achieve the same SNR. This difference, in dB is:

$$10 \log\left(\frac{12}{9}\right) = 1.25 \text{ dB}$$

Answers

To achieve the same SNR from the same transmitted power, the length of the links should result in a power drop difference equal to the difference above.

Link A indeed provides a smaller attenuation by 0.02 dB/km, therefore, to compensate for the above difference, the link length should be:

$$L = \frac{1.25\ \text{dB}}{0.02\ \text{dB/km}} = 62.5\ \text{km}$$

b) The bitrate is estimated assuming a transmitted power of 10 mW (10 dBm)
At link A, the received power will be:

$$I_R = 10\ \text{dBm} - 0.45\frac{\text{dB}}{\text{km}} \cdot 62.5\ \text{km} = 10\ \text{dBm} - 28.125\ \text{dB} = -18.125\ \text{dBm}$$

Hence: $I_R = 15\ \mu\text{W}$

The SNR will be:

$$\text{SNR} = \frac{I_R}{\text{NEP} \cdot \sqrt{\Delta f}}$$

Taking SNR=12, :

$$\Delta f = \left(\frac{15 \cdot 10^{-6}}{12 \cdot 12 \cdot 10^{-12}}\right)^2 = 10\ \text{GHz}$$

Hence the bit rate will be:

$$B = 2 \cdot \Delta f = 20\ \text{Gbps}$$

Using a rough indicative responsivity of $R = 1\ \text{A/W}$, a shot noise dominated SNR would be:

$$\text{SNR}_{\text{SHOT}} = \sqrt{\frac{I_R \cdot R}{e \cdot B}} = 68.4$$

This SNR is much larger than that of the receiver (SNR = 12). Therefore, the shot noise is 68.4/12 less than the receiver noise, which demonstrates that the receiver noise dominates.

c)
The SNR from shot noise and receiver noise will respectively be:

$$\text{SNR}_{\text{SHOT}} = \sqrt{\frac{I_R \cdot R}{e \cdot B}}$$

Answers

$$SNR_R = \frac{I_R}{NEP \cdot \sqrt{\Delta f}}$$

Equating the two:

$$\frac{\mathrm{I_R} \cdot \mathrm{R}}{\mathrm{e} \cdot \mathrm{B}} = \frac{I_R^2}{NEP^2 \cdot \Delta f}$$

And hence:

$$2 \cdot I_R = NEP^2 \cdot \frac{R}{e} = NEP^2 \cdot \frac{I_e}{e\,\Phi} = NEP^2 \cdot \frac{N_e}{N_{ph} \cdot E_{ph}} = NEP^2 \cdot \frac{\eta_Q}{E_{ph}} = \frac{NEP^2}{E_{ph}}$$

d)
The SNR will be shot dominated, hence:

$$\mathrm{SNR_{SHOT}} = \sqrt{\frac{\mathrm{I_R} \cdot \mathrm{R}}{\mathrm{e} \cdot \mathrm{B}}}$$

$$\mathrm{SNR_{SHOT}^2} = \frac{\mathrm{I_R} \cdot \frac{\mathrm{I_e}}{\mathrm{I_R}}}{\mathrm{e} \cdot \mathrm{B}} = \frac{\mathrm{N_e} \cdot \frac{\mathrm{e}}{\mathrm{t}}}{\mathrm{e} \cdot \frac{\mathrm{N_{BIT}}}{\mathrm{t}}} = \mathrm{N_e/N_{BIT}}$$

This ratio provides the number of electrons contributing to each bit of information received.

5. A silicon p-i-n photodiode has *p*, *i* and *n* doping levels respectively of N_A, $N_D^- = 10^{20}\ m^{-3}$ and N_{D^+}. The distances from the diode surface to the top and bottom of the intrinsic layer are $X_1 = 1.0$ μm and $X_2 = 5$ μm, respectively.

a) Find the optical attenuation coefficient that maximises the fraction of incident photons which are absorbed in the intrinsic layer of this device.

[4]

b) Find the value of N_{D^+} that allows the peak electric field magnitude to be only 15% greater than the magnitude at the bottom of the intrinsic layer, while only depleting 0.4 μm of the *n* layer. Then, find the value of N_A that allows these conditions to be achieved while only depleting 0.4 μm of the *p* layer.

[8]

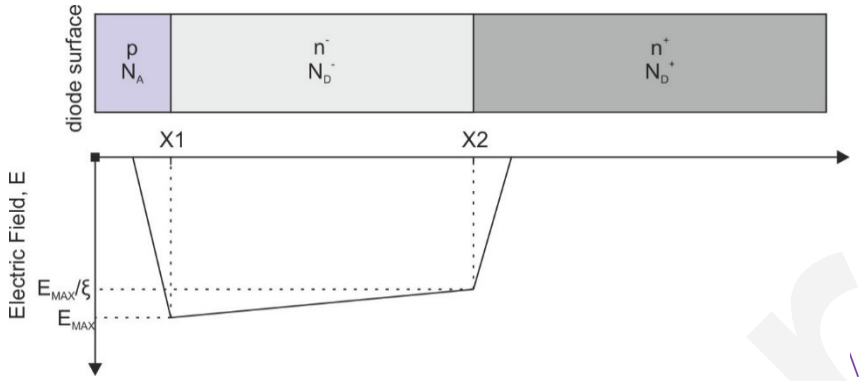
c) Find the applied bias voltage *V* such that the conditions in (b) are achieved and the peak electric field magnitude is 2.5×10^5 V/m.

[4]

d) Discuss why it may be beneficial to minimise the distances to which the depletion region extends into the *n* and *p* layers.

[4]

Solution:



a) Neglecting reflections, the fraction of photons absorbed in the intrinsic region will be:

$$\eta = e^{-aX1} - e^{-aX2}$$

To find the value of a that maximises η:

$$\frac{d\eta}{da} = -X1e^{-aX1} + X2\, e^{-aX2}$$

Setting $d\eta/da = 0$:

$$\frac{X1}{X2} = e^{-a(X2-X1)}$$

$$a = \frac{\ln\left(\frac{X1}{X2}\right)}{X1 - X2} = 0.1733\ \mu m^{-1}$$

b) In the p, the intrinsic and the n region, the slope of electric field will be:

$$E_{x,p} = \frac{\partial E}{\partial x} = \frac{\rho_p}{\varepsilon} = \frac{qN_A}{\varepsilon}$$

$$E_{x,i} = \frac{\partial E}{\partial x} = \frac{\rho_i}{\varepsilon} = -\frac{qN_D^-}{\varepsilon}$$

$$E_{x,n} = \frac{\partial E}{\partial x} = \frac{\rho_n}{\varepsilon} = -\frac{qN_D^+}{\varepsilon}$$

The maximum field occurs at the p/i interface:

$$E(X1) = E_{MAX}$$

But E_{MAX} is $\xi = 1.1$ larger than E(X2):

$$E_{MAX} = \xi \cdot E(X2)$$

Hence:

$$E_{x,i} = \frac{E(X2) - E(X1)}{X2 - X1} = \frac{E_{MAX}\left(\frac{1}{\xi} - 1\right)}{X2 - X1} = E_{MAX}\frac{(1-\xi)}{\xi} \cdot \frac{1}{X2 - X1}$$

And:

$$E_{x,n} = \frac{0 - E(X2)}{w_n} = -E_{MAX} \cdot \frac{1}{\xi} \cdot \frac{1}{w_n}$$

Therefore:

$$\frac{N_D^+}{N_D^-} = \frac{E_{x,n}}{E_{x,i}} = \frac{-\frac{1}{w_n}}{\frac{1-\xi}{X2 - X1}} = \frac{1}{\xi - 1} \cdot \frac{X2 - X1}{w_n} = 67$$

And

$$N_D^+ = 6.7\ 10^{21}\ m^{-3}$$

In the p region:

$$E_{x,p} = \frac{E(X1)}{w_p} = \frac{E_{MAX}}{w_p}$$

Therefore:

$$\frac{N_A}{N_D^-} = -\frac{E_{x,p}}{E_{x,i}} = \frac{-\frac{1}{w_p}}{\frac{(1-\xi)}{\xi} \cdot \frac{1}{X2 - X1}} = \frac{\xi}{\xi - 1} \cdot \frac{X2 - X1}{w_p} = 76.7$$

And

$$N_A = 7.6\ 10^{21} m^{-3}$$

c)

In the n region:

$$E_{x,n} = -\frac{E(X2)}{w_n} = -\frac{E_{MAX}}{\xi w_n}$$

$$V = \int_{X1-wp}^{X2+wn} E dx =$$

Answers

$$= \int_{X1-wp}^{X1} \frac{E_{MAX}}{w_p}(x - X1 + w_p)dx$$

$$+ \int_{X1}^{X2} E_{MAX}\left(1 + \frac{(1-\xi)}{\xi} \cdot \frac{x - X1}{X2 - X1}\right)dx + \int_{X2}^{X2+w_n} \frac{\mathrm{E_{MAX}}}{\xi \mathrm{w_n}}(\mathrm{x_2} + \mathrm{w_n} - \mathrm{x})\mathrm{dx}$$

$$= \frac{1}{2}\mathrm{E_{MAX}w_p} + \frac{1}{2}\mathrm{E_{MAX}} \cdot \frac{\xi + 1}{\xi}(\mathrm{X2} - \mathrm{X1}) + \frac{1}{2}\mathrm{E_{MAX}} \cdot \frac{1}{\xi}\mathrm{w_n}$$

I.e.:

$$\mathrm{V} = \mathrm{E_{MAX}}\left(0.25 + \frac{2.1}{1.1}2.5 + \frac{0.25}{1.1}\right) = 4.11 \cdot 10^{-6} \cdot \mathrm{E_{MAX}}$$

$$\mathrm{V} = 4.11 \cdot 10^{-6} \cdot 2.5 \cdot 10^{5}\ \mathrm{V} = 1.03\ \mathrm{V}$$

d)

Keeping the depletion region thin provides the following advantages:

- The depletion region width and the capacitance are less dependent on voltage.
- The electric field is much more uniform
- Keeps a conductive path in p which is needed for circuit current flow.

Answers